ENHANCED SCIENCE 8

EARL CRIS B. RIGGS
TEACHER

ENHANCED SCIENCE Q1 W1
LESSON: FORCES

Learning Objective:

Investigate the relationship between the amount of force applied and the mass of the object to the amount of change in the object's motion.
(Week 1 S8FE-Ia-15)

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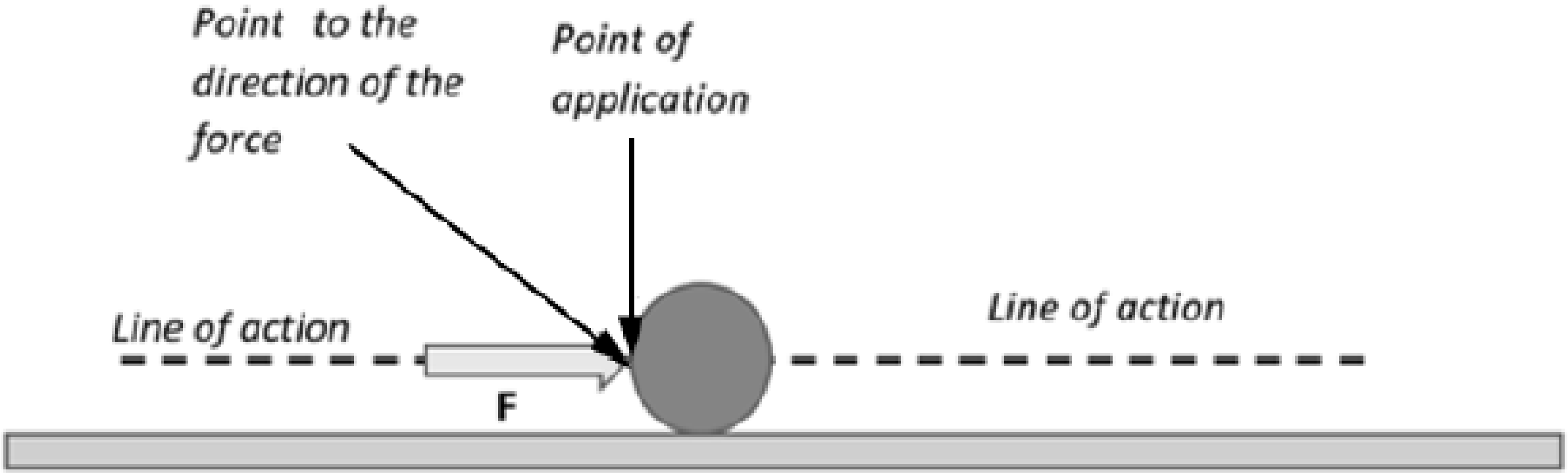

Figure 7. Physical concepts illustrated on a ball

Illustrated by: Rhenan O. Bacolod

Figure 7 shows how force acts on a ball, but you need to be familiar with the following terms:

- **magnitude** – refers to the size or strength of the force. It is commonly expressed in Newton (N) for Meter-Kilogram-Second (MKS) system, Dyne (dyn) for Centimeter–Gram–Second (CGS) system and pounds (lbs) for Foot–Pound–Second (FPS) system. In the International System of Units (SI), Newton is commonly used which is named after Sir Isaac Newton, an English physicist and mathematician.
- **direction** – points to where the object goes. The direction of the arrowhead indicates the direction of the force. The length of the arrow represents the amount of force (relative magnitude).
- **point of application** – the location of where the force is applied.
- **line of action** – is the straight line passing through the point of application and is parallel to the direction of force.

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There are two types of force, namely:

Contact forces – forces where objects touch or contact with each other.

Examples of contact forces:

1. **Applied** – a force given to a person or object by another person or object. Its symbol is **F** depending on who or what applies force to the object. If a boy applies a force to a wall, we denote it with $\mathbf{F_{BOY}}$. Refer to the figure below.

Figure 8. Applied force of a boy ($\mathbf{F_{BOY}}$) towards the wall

2. **Friction** – is the force acting against or opposite an object in contact with which makes the movement of the object slow down. Friction always opposes the motion of an object. Its symbol is written as $\mathbf{F_f}$. ***Air resistance*** denoted by $\mathbf{F_{AIR}}$ is an example of frictional force of the air against a flying kite, airplanes, parachutes or those in skydiving sports. For free-falling objects, this force is always considered negligible, meaning the magnitude is unnoticeable.

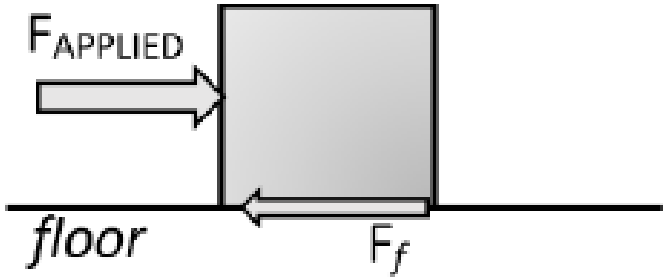

Figure 9. Frictional force ($\mathbf{F_f}$) acts opposite to the force applied

3. **Normal** – is the force that acts perpendicular to the surface of the object in contact with. Its symbol is $\mathbf{F_N}$.

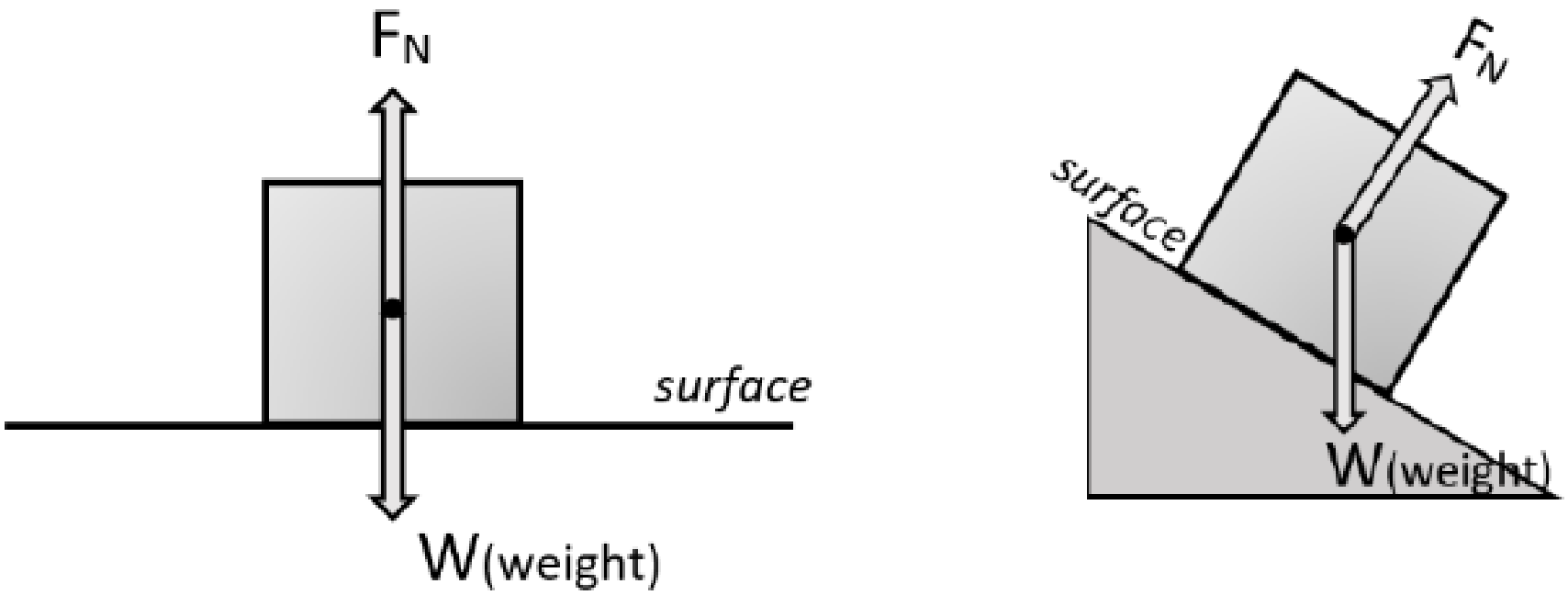

Figure 10. Normal force ($\mathbf{F_N}$) acts perpendicular to the surface

4. **Tension** – is the force applied to string, rope, chain or cable. Its symbol is **T**.

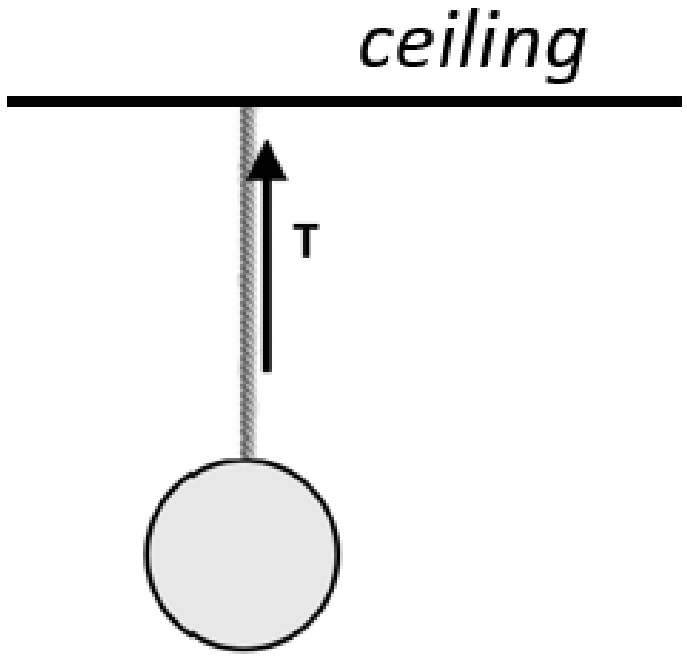

Figure 11. Tensional force or simply tension (**T**) that acts on the rope of a hanging ball.

Illustrated by: Rhenan O. Bacolod

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Non-contact forces – forces where objects do not touch or contact with each other. These forces act over a zone or area called field.

Examples of non-contact forces:

1. **Gravitational (F_g)** – is the force of attraction between two objects. In the case of the Earth, this gravitational force causes objects to fall down to the ground. It makes satellites and smaller objects stay in orbit near the more massive planets. Mass and distance of the two objects affect the gravitational force that holds them. The bigger the masses of the objects are, the bigger is the gravitational force between them. The closer the objects are, the greater is the gravitational force between them. The figure below illustrates gravitational force between the Earth and the Moon. Earth has bigger gravitational force over the Moon.

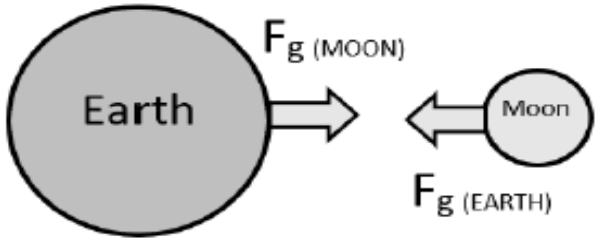

Figure 12. Gravitational force between the Earth and the Moon.

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The weight of an object, denoted by **W**, is an example of the gravitational force of the Earth towards the object. In figure 13, the weight of a book **(W_{BOOK})** is illustrated.

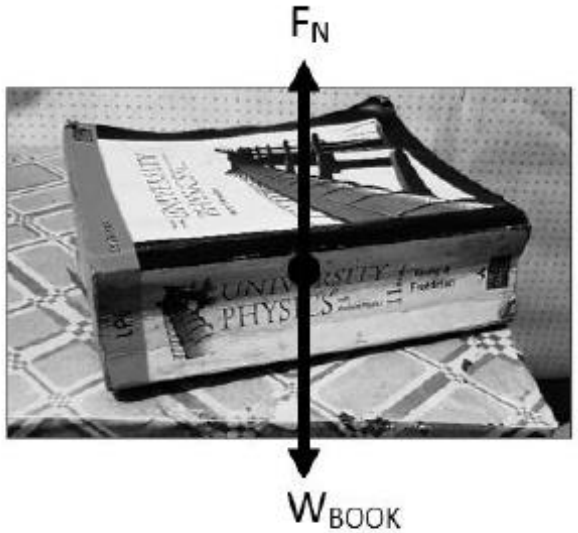

Figure 13. Illustration of the weight of a book **(W_{BOOK})** on the table with normal force (F_N)

Photo credit: Rhenan O. Bacolod

However, the weight of an object depends on the mass of the celestial body where the object is attracted to. Meaning, we seem to be lighter when we are on the moon than on the Earth.

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2. **Magnetic–** are forces exerted on a field of attraction or repulsion as in the case of magnets and other magnetic materials. Magnets and magnetic materials have two poles – the north and south poles. Attraction may occur when two poles are not the same, a positive and a negative while repulsion takes place with the same poles, positive-positive and negative-negative.

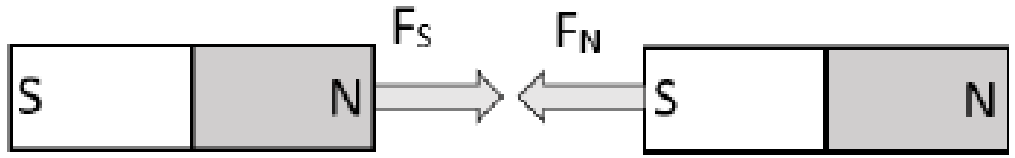
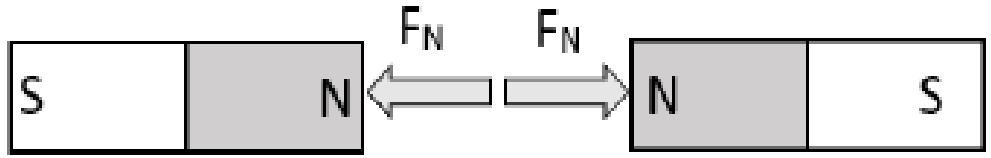


Figure 14. Opposite poles attract



Like poles repel

Illustrated by: Rhenan O. Bacolod

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Balanced Forces

To describe a force, you must know two things. You must know the magnitude and the direction of the force. Suppose two teams are playing tug-of-war as shown in figure 15. Each team is pulling with equal magnitude of force, F_A and F_B , on the rope but in opposite directions. Neither team can make the other team move. Forces that are equal in magnitude but opposite in direction are called ***balanced forces***. Balanced forces do not cause a change in motion. When balanced forces act on an object at rest, the object will not move.

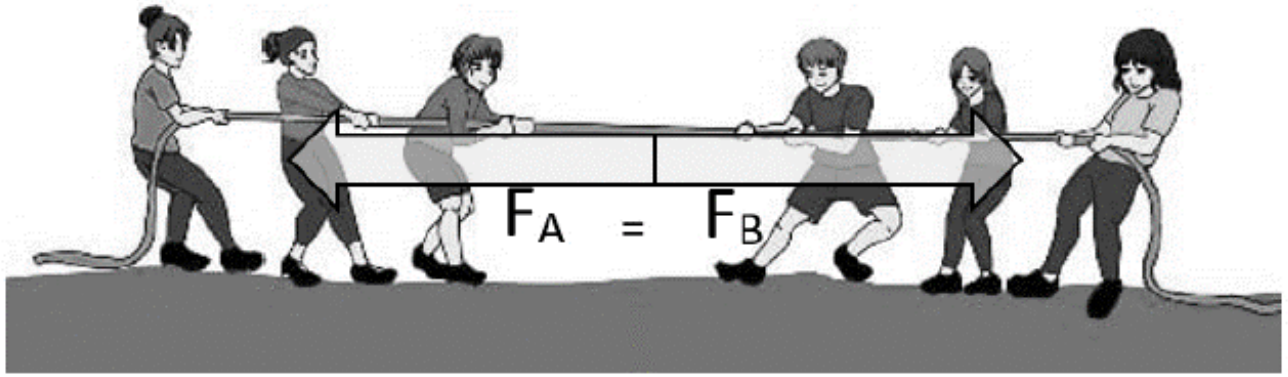

Figure 15. A representation of balanced forces in a tug-of-war game

Illustrated by: Rosa Mia L. Pontillo

Unbalanced Forces

When you push a table and then it moves, unbalanced forces are present. Forces that cause a change in the motion of an object are ***unbalanced forces***. Unbalanced forces are not equal and in opposite direction. Suppose that one of the teams in tug-of-war, as shown in figure 16, exerts greater magnitude of force, F_B, on the ground than the other team, the forces applied on the ground would no longer be equal. One team would be able to pull the other team in the direction of the larger force.

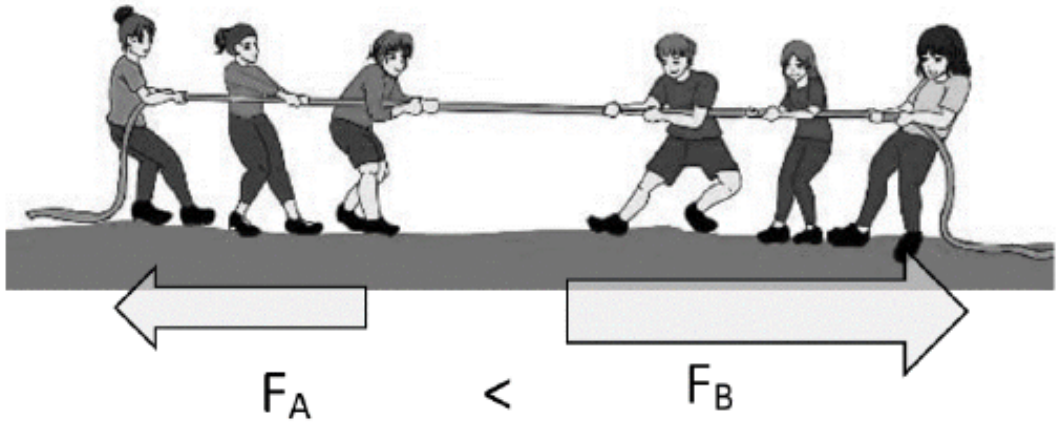

Figure 16. A representation of unbalanced force in a tug-of-war game

Illustrated by: Rosa Mia L. Pontillo

Net or Resultant Force

In an object, there may be several forces acting on it. ***Net force*** or ***resultant force*** is the sum of all forces acting on an object. Two or more forces in the same line of action exerted on an object are balanced if their effects cancel each other.

When an object is at rest, a zero net force would make the object remains at rest. Moreover, when the object is moving, a zero net force would make the object maintain its velocity at a given time interval.

On the other hand, when the net force is not zero, the object's velocity will change. A net force exerted to an object at rest may cause the object to move. In a moving object, a net force will increase its velocity when the force is in the same direction of its motion. If the net force is in the opposite direction of the object's motion, the force will reduce the object's velocity. When the net force acts sideways on a moving object, the direction of the object's velocity changes.

ACTIVITY TIME!

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Procedure:

Examine the pictures below:

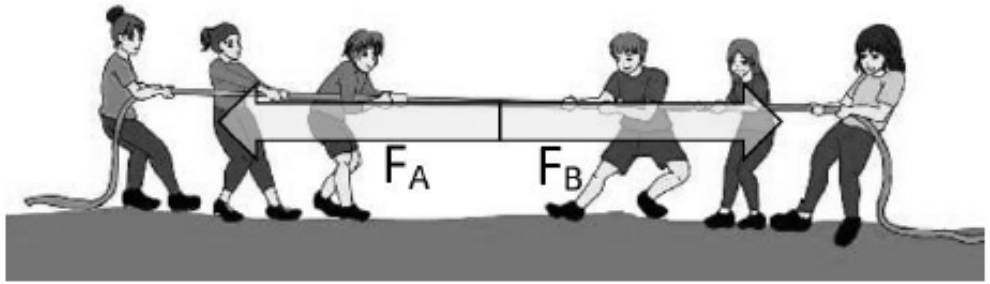

Figure 18. Balanced forces in a tug-of-war.
Illustrated by: Rosa Mia L. Pontillo

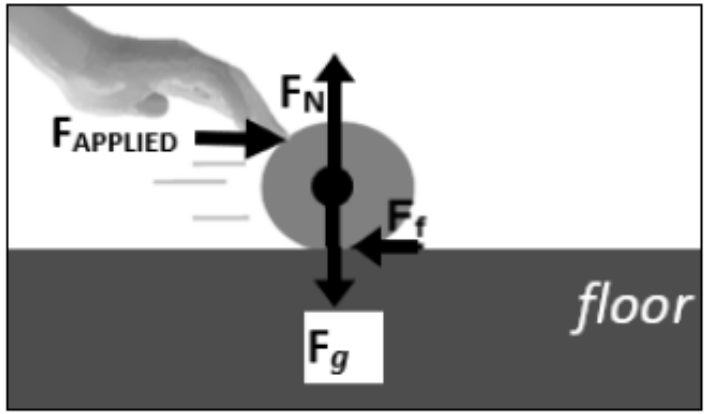

Figure 19. A ball pushed on the floor
Illustrated by: Rosa Mia L. Pontillo

F_A - applied force on the rope by Team A
F_B - applied force on the rope by Team B

F_N - force exerted by the floor on the ball
F_g - force exerted by the Earth on the ball
F_f - opposite force in the direction of the motion or frictional force
$F_{APPLIED}$ – applied force on the ball

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For question numbers 1-3, refer to Figure 18.

1. If two groups in the game are exerting equal and opposite forces on the rope, will the rope move?

   _____

2. In the same condition, what happens to the net force acting on the rope?

   _____

3. What are the forces present on the rope?

   _____

4. Refer to Figure 19. If you place a ball on the floor then push it gently to one side, observe the motion of the ball as it rolls down the floor.

   a. What makes the ball stop rolling after sometime?

   _____

   b. What are the forces that act on the ball?

   _____

ACTIVITY TIME!

Activity 4. Unbalanced forces

Objectives:

After performing this activity, you should be able to:

1. Examine the situations when the forces are unbalanced; and
2. Explain the effect of unbalanced forces on the state of motion of an object.

Materials:

Pictures

Procedure:

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Procedure:

Examine the pictures below.

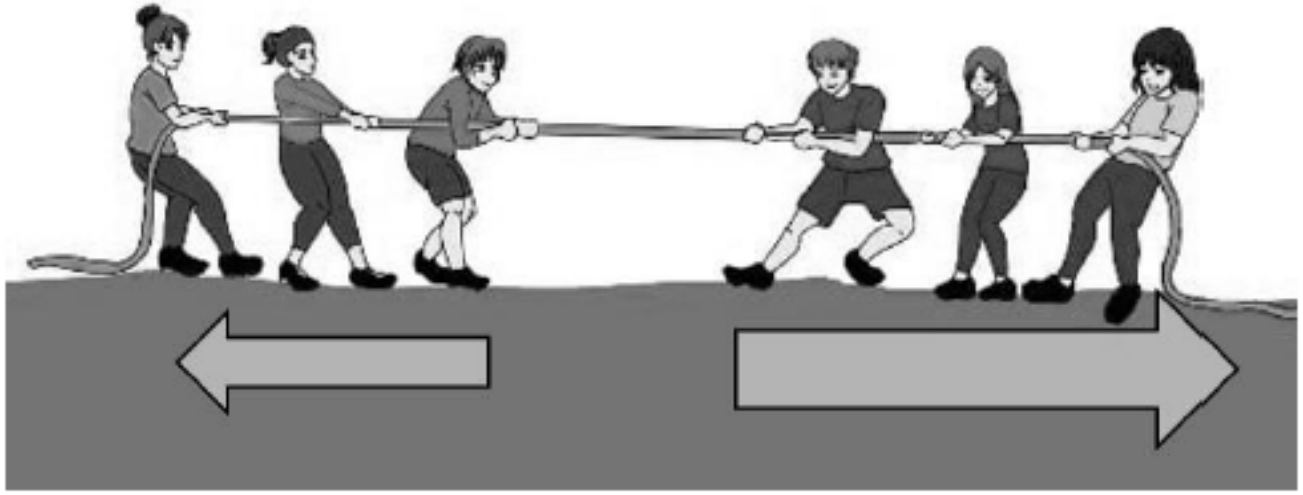

Fig 20. Unbalanced forces in a tug-of-war

Illustrated by: Rosa Mia L. Pontillo

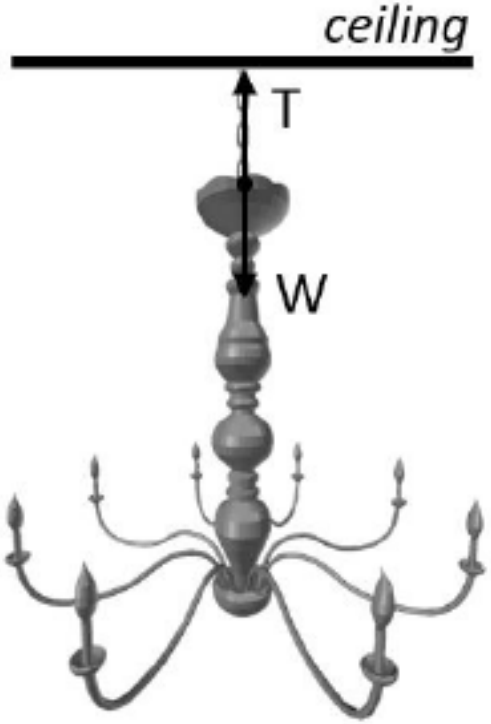

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Fig 21. A chandelier hangs on a chain.

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For question numbers 1-2, refer to figure 20.

1. If two groups in the game are exerting unequal and opposite forces on the ground, will the rope move?

   _____

2. Will the net force be equal to zero?

   _____

3. To what direction does the rope move?

   _____

For question numbers 4 – 6, refer to figure 21.

4. Are the forces acting on the chandelier unbalanced?

   _____

5. What force can cause the chandelier to fall?

   _____

6. What type of force is your answer in number 4?

   _____

SHORT QUIZ!

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Fill in the blanks with correct word/s. Write your answers on a separate sheet of paper.

1. Any push or pull is called _____.
2. _____ refers to the size or strength of the force while _____ points to where the object goes. The direction of the arrowhead indicates the direction of the force. The length of the arrow represents the amount of force (relative magnitude).
3. There are two types of forces, namely _____ and _____.
4. Applied, friction, normal and tensional forces are examples of _____ forces.
5. Gravitational and magnetic forces are examples of _____ forces.
6. _____ are forces that are equal in magnitude but opposite in direction.
7. _____ are forces that cause a change in the motion of an object.
8. The sum of all forces acting in an object is called _____.
9. If the forces in an object are balanced, the net force is _____.
10. If the forces in an object are unbalanced, the net force is _____.

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Fill in the blanks with correct word/s. Write your answers on a separate sheet of paper.

1. Any push or pull is called ___FORCE___.
2. ___MAGNITUDE___ refers to the size or strength of the force while ___DIRECTION___ points to where the object goes. The direction of the arrowhead indicates the direction of the force. The length of the arrow represents the amount of force (relative magnitude).
3. There are two types of forces, namely ___CONTACT___ and ___NON-CONTACT___
4. Applied, friction, normal and tensional forces are examples of ___CONTACT___ forces.
5. Gravitational and magnetic forces are examples of ___NON-CONTACT___ forces.
6. ___BALANCED___ are forces that are equal in magnitude but opposite in direction.
7. ___UNBALANCED___ are forces that cause a change in the motion of an object.
8. The sum of all forces acting in an object is called ___NET OR RESULTANT FORCE___.
9. If the forces in an object are balanced, the net force is ___ZERO___.
10. If the forces in an object are unbalanced, the net force is ___NOT ZERO___.

When we combine or add forces to determine the net or resultant force, we will limit to those forces which act along the same line of action. The algebraic signs + and – are used to indicate the direction of forces. Unlike signs are used for forces acting in opposite directions, see figure 22 below.

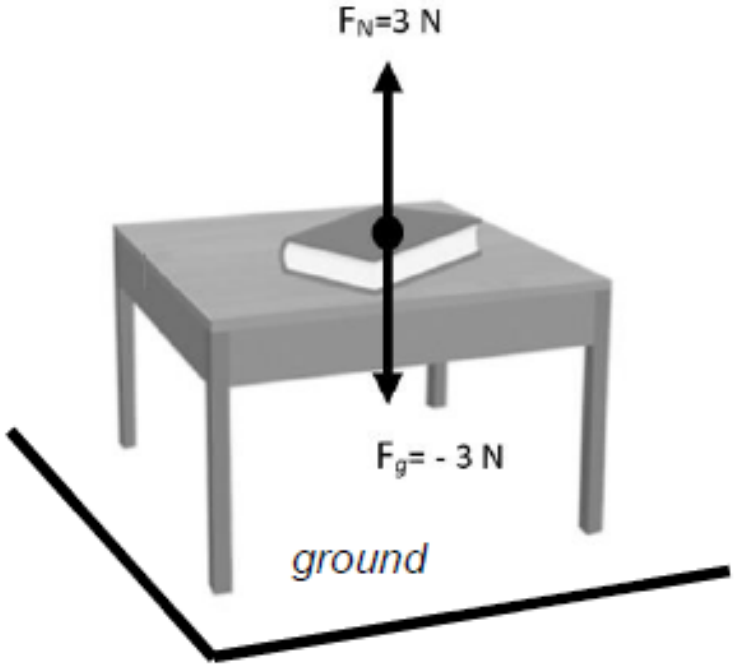

Figure 22. Forces in opposite direction
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The force of gravity (F_g) and the normal force (F_N) are assigned opposite signs. F_N is given a positive (+) sign while F_g is given a negative (-) sign. If both F_g and F_N are given a magnitude value of 3 N, then the net force along this line (vertical) is represented by $\sum F$.

To solve for the net force, you get the sum of all the forces exerted in the object. The symbol sigma, $\sum$, means you add all the existing forces in the same line of action. You cannot add a vertical force to a horizontal force. In Figure 20, we can add the two forces, normal and gravitational force since they are in the same vertical line of action.

$$F_{net} = \sum F$$

$$F_{net} = F_N + F_g$$

$$= 3\ N + (-3\ N)$$

$$= 0\ N$$

If the net force equates to zero, they are considered balanced. If the net force is not equal to zero, the forces are not balanced. This unbalanced force would cause a change in a body's state of motion. Refer to the figure 23.

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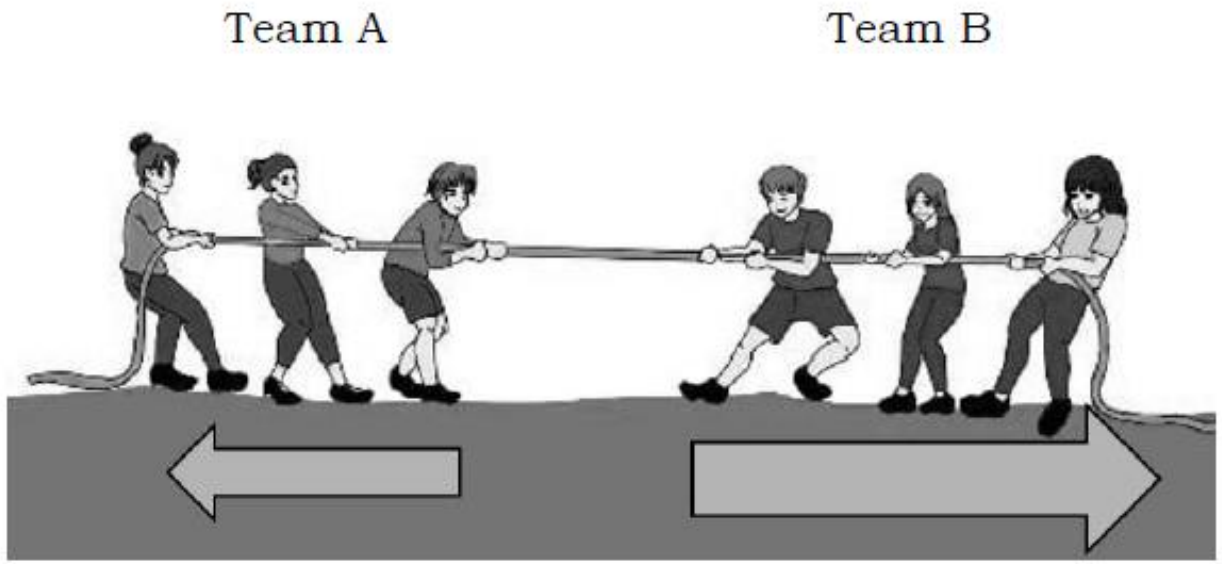

Figure 23. Forces in opposite direction.
Illustrated by: Rosa Mia L. Pontillo

In figure 23, suppose one side in the tug-of-war, team A exerts 1,000 N while the other side, Team B exerts 1300 N. To get the net force,

$$F_{net} = F_A + F_B$$

$$= (-1000\ N) + 1300\ N$$

$$= +300\ N \text{ or } 300\ N, \text{ to the right}$$

LET'S SOLVE!

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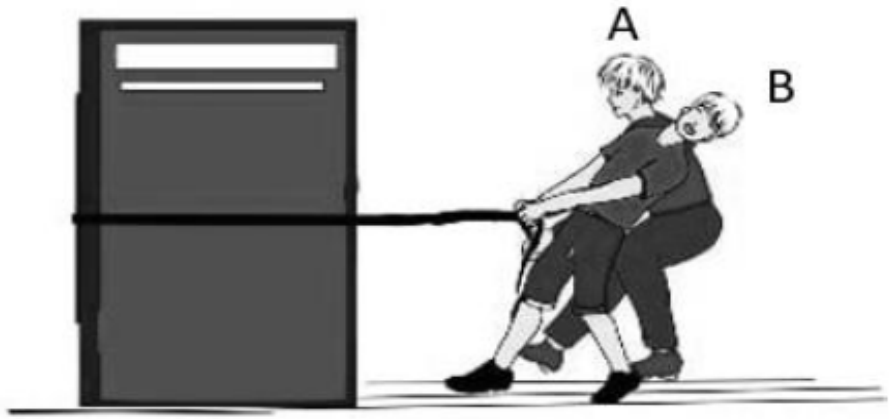
Now, try to answer this concept check in the situations below.

Read and analyze each problem carefully. Write your answers and solutions in your answer sheet.

1. Boys A and B, are pulling a heavy cabinet at the same time with 5 N of force each. What is the net force acting on the cabinet? _____



Illustrated by: Rosa Mia L. Pontillo

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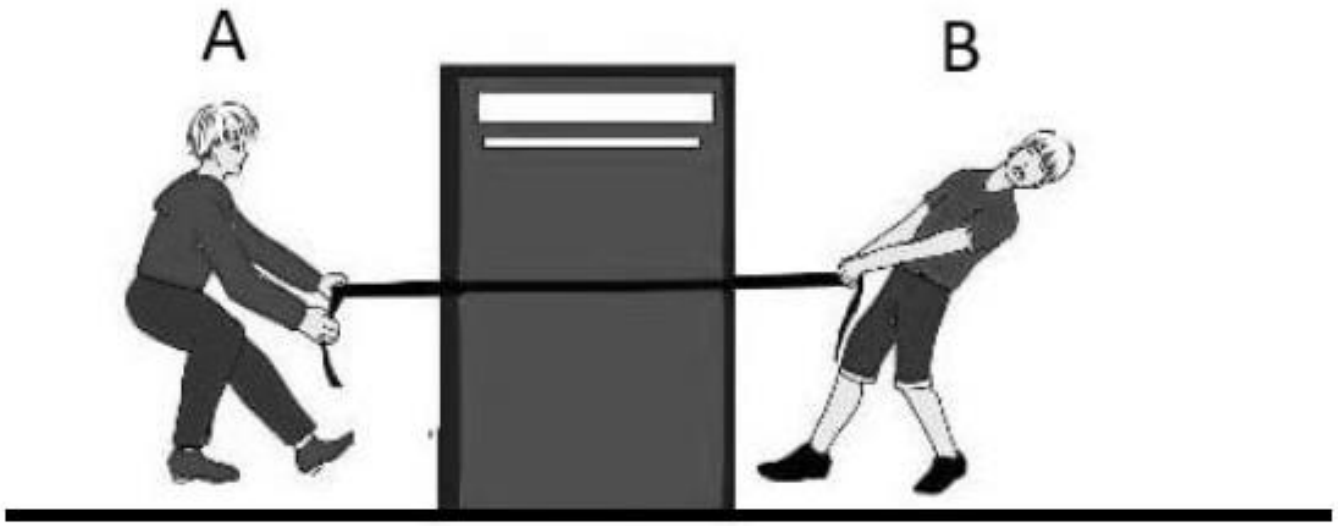
2. What if boy A and boy B pull the heavy cabinet at the same time in opposite directions with 10 N and 5 N of force respectively,
   a. what will be the net force on the cabinet? _____
   b. will the cabinet move? _____
   c. to what direction will it move? _____



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3. From problem 2, suppose another boy, Boy C pulls the heavy cabinet with 5 N of force in the same direction with Boy A,
   a. What will be the net force on the cabinet? _____
   b. Will the cabinet move? _____
   c. In what direction will the cabinet move? _____

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ASSESSMENT TIME!

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Choose the letter of the correct answer. Write your answers on a separate sheet of paper.

1. Which is NOT a unit of force?
   A. Dyne
   B. Joules
   C. Newton
   D. Pound

2. What refers to how large or strong is the force?
   A. magnitude
   B. line of action
   C. velocity of the object
   D. direction of the force

3. What force is always directed opposite to the motion of an object?
   A. Friction
   B. Gravitational
   C. Magnetic
   D. Tension

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4. What force pulls an object back to the Earth?
   A. Applied
   B. Gravitational
   C. Magnetic
   D. Tension

5. Which surface would be easiest for a bicycle to move?
   A. Sand
   B. Grass
   C. Muddy road
   D. Concrete road

6. What type of forces holds between two celestial bodies like the Sun and Earth?
   A. Applied
   B. Gravitational
   C. Magnetic
   D. Tension

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7. What force acts perpendicular to the surface of the object in contact with?
   A. Applied
   B. Gravitational
   C. Normal
   D. Tension

8. Two people pull on a rope in a tug-of-war. Each pull with a 300 N force. What is the net force?
   A. 0 N
   B. 300 N
   C. 600 N
   D. 900 N

9. Planet Jupiter has a bigger mass than the Earth. If you were able to travel to planet Jupiter, what happens to your weight?
   A. increases
   B. decreases
   C. stays the same
   D. varies with day and night

10. What is the net force in the figure below?

Illustrated by: Rhenan O. Bacolod

A. 25 N, to the left
B. 25 N, to the right
C. 195 N, to the left
D. 195 N, to the right

ASSIGNMENT

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Analyze and answer the problems below. Write your answers on a separate sheet of paper.

1. The wheel barrow at rest containing stones has a force of 1000 N. A man exerted a push of 500 N.

   a. Will he able to move the wheel barrow? _____

   b. Suppose another man tried to help him and exerted a force of 500 N. Will they able to move the wheel barrow?

   _____

   c. Another man saw their struggle in pushing the wheel barrow. He tried to help them. He exerted a force of 400 N. Will the three of them move the wheel barrow?

   _____

2. A box is at rest and a force of 250 N is applied to it. If the frictional force is 200 N, determine the net force.

TOPIC NEXT WEEK

WEEK 2:
LAWS OF MOTION